Note:~ We can take member as private when using member function.

this pointer :

The this pointer is often used inside a constructor to initialize member variables when parameter names are same as member variable names.

Eg;

```
~~#include <iostream>~~
#include<iostream>
using namespace std;
class student
 {
    public :
     // constructor using 'this' pointer
    int age;
   student( int age)
     {
       this->age = age //resolving naming conflicts
     }
 };

int main()
{
   student s(20);
   cout<<" age="<<s.age;
}
```

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→ Use:-

① Resolves Name conflict → this→name = name; ensure the member variable gets the correct value.

② Makes code readable → Clearly distinguishes between member variables and parameter.

③ Used for method chaining → When returning *this, it enables function chaing.

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③ Copy Constructor

A copy constructor in C++ is a special constructor that creates a new object as a copy of an existing object.
It is used to initialize an object using another object of the same class.

Syntax:

```
Class class_name
  {
      public:
      class_name ( class_name &new_object_name)
      {
          // copy constructor body
      }
  };
```

Akash

```
// C++ program for copy constructor

#include<iostream>
using namespace std;
class student
{
    public :
     int age;

    // parameterized constructor

    student(int age)
    {
        this->age = age; // using 'this' pointer
    }

    // copy constructor

    student(student &s)
    {
        age = s.age;
    }
};
int main()
{
    Student s1(20); // creating object
    student s2 = s1 ; // copy constructor is called
    cout << " age= " << s2.age;
}
```

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// C++ program for copy constructor using member function

```
#include<iostream>
using namespace std;
class student
  {
      private :
        int age;
      public :
        student(int age)
        {
          this→ age = age;
        }
        // copy constructor
        student (student &s)
        {
            age = s.age
        }
        void show()    // member function
        {
            cout << "age = " << age;
        }
};
    int main()
{
        student s1(20);
        student s2 = s1; // copy constructor is called
        s2.show; // function call
}
```

Akash